Radar & AI for Human Computer Interaction via Hand Gestures

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Summary

This report classifies four radar hand gestures (wave, pinch, swipe and click) from the micro-Doppler spectrograms of the DopNet dataset: six persons, 40 samples per gesture and person. A test set of 20% is separated first. Each spectrogram is pre-processed separately: the noise is removed with a threshold 10 dB above the noise level of the far Doppler bins, the Doppler axis is cropped and the values are scaled to $[0, 1]$. The models use either the spectrograms resized to 128×128 pixels or the micro-Doppler envelope features of Amin et al. Logistic regression, an SVM and a random forest are trained on the features, and an SVM on principal components and a convolutional neural network (CNN) on the images. Models and hyperparameters are chosen with 5-fold cross-validation on the training set, and leave-one-subject-out (LOSO) shows how they generalise to a new person. A CNN with five convolution blocks is the final model, with a validation accuracy of 0.967 and a test accuracy of 0.948 (macro-F1 0.948). For a new person its accuracy drops to 0.74, the level of the feature models, so more persons are needed to recognise a gesture regardless of who performs it.

1 Machine Learning Pipeline

The data comes from the DopNet dataset published by the UCL Radar Group[1]. It has been recorded using an Ancortek 24 GHz FMCW radar with a bandwidth of 750 MHz. The recorded samples are micro Doppler spectrograms that show the radar magnitude as a function of time and Doppler frequency. The Doppler frequency is proportional to the radial velocity of the measured moving object. Movement towards the radar generates a positive Doppler shift, while movement away from the radar generates a negative Doppler shift.

The used dataset contains 960 labelled samples of different gestures made with a human hand. Each sample is provided as a CSV file, labelled with its gesture, person and repetition. There are 4 gestures (wave, pinch, swipe and click) performed by 6 persons. The measurement for each combination has been repeated 40 times, resulting in a perfectly balanced dataset.

Each sample has 800 Doppler bins, where zero Doppler is in the centre at bin 400. The dataset does not contain NaN or infinite values. The magnitude values have already been normalized to each sample's maximum at 0 dB. The goal of this project is to train a model that can classify the 4 gestures based on their spectrograms. The main model will focus on recognising gestures only from people it has seen before. The generalization of the chosen models to new people will also be tested, but that is not the primary objective for this report.

1.1 Pipeline

Figure 1 gives an overview of the pipeline. First, the 960 samples are split into a training set of 768 samples and a test set of 192 samples (section 1.2). Every spectrogram is then pre-processed separately through noise removal, cropping and scaling (section 2).

The completely pre-processed spectrograms are used for the image models: a convolution neural network (CNN) and an SVM on principal components (SVM + PCA). The spectrograms without noise are used to compute envelope features using the paper by Amin et al (section 2.3). In total 760 features are computed. The set of features a model uses (3, 44 or 716) is treated as a hyperparameter. Standardization and PCA are fitted on the training folds only.

The hyperparameters of every model are tuned with a grid search on stratified 5-fold cross-validation on the training set. For every model the setting with the highest mean validation accuracy is chosen, and the final model is the model with the highest validation accuracy overall. These settings are then fixed. Leave-one-subject-out (LOSO) on the training set shows how each model generalises to a new person. Finally, every model is retrained on all 768 training samples and evaluated once on the test set. The models are compared on accuracy, macro-averaged F1 score and confusion matrices (section 3).

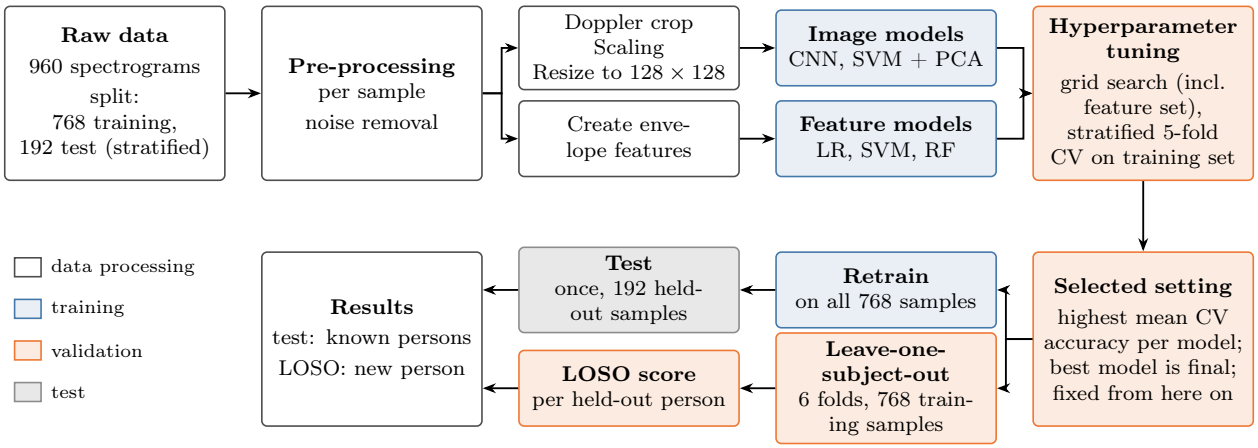


Figure 1: Overview of the machine learning pipeline.

[1]UCL Radar Group, "DopNet radar hand-gesture dataset," `https://github.com/UCLRadarGroup/DopNet`.

1.2 Data split

Before any further data exploration or processing, the data is split into three roles: training, validation and test. This guarantees that choices will not be made based on test data. A test set of 20% (192) samples is separated as shown in Figure 2. The split is stratified on person and gesture, so every combination has the same share in the test set. This keeps the training and test data balanced. The test set will not be touched until the final evaluation of the model.

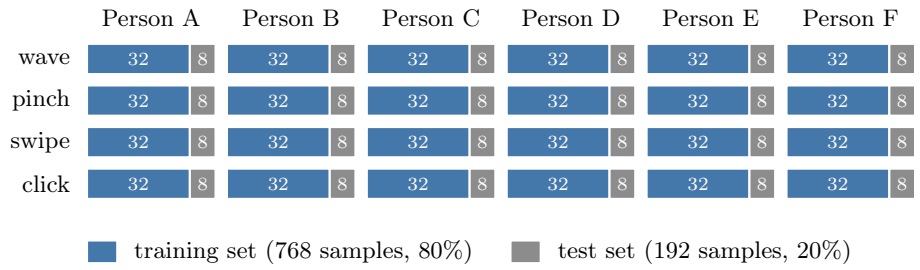
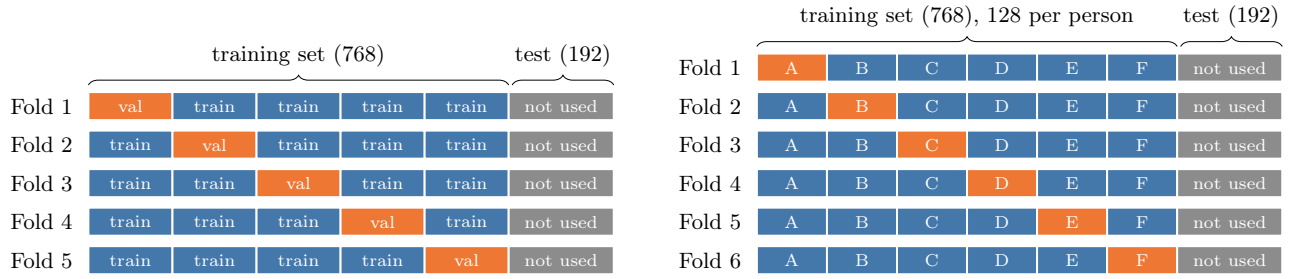


Figure 2: Stratified hold-out test set.

The training set contains the remaining 80% (768) of the data. Every choice that is learned from data will only use this set. This includes the data exploration, noise parameters, feature choices and hyperparameter selection. Validation is done with stratified 5-fold cross validation on the training set, as shown in Figure 3a. The training set is split into 5 balanced folds. Each model is trained on 4 folds and then evaluated on the remaining 5th fold. This is repeated 5 times so that each fold acts as validation set exactly once. The model and hyperparameters will be picked based on the highest mean validation accuracy.



(a) Stratified 5-fold cross-validation: hyperparameter and model selection

(b) Leave-one-subject-out: generalisation to a new person

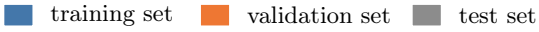


Figure 3: Validation folds.

After hyperparameter and model selection, the chosen models will be retrained on all 768 training samples. The final model is then evaluated once using the 192 test samples. The model performance measures how well the gestures of known persons are recognised.

This test score will likely be higher than the accuracy for a new person that the model has not seen during training. To estimate the generalization to unseen people, the same models with the already fixed hyperparameters will be retrained using Leave-one-subject-out (LOSO). The model trains on five persons and tests on the 6th unseen person, as shown in Figure 3b. Since the hyperparameters were selected based on all people, the accuracy will be slightly optimistic. However, it is beyond the scope of this report to retune the hyperparameters with a nested LOSO split. Instead, this step just acts as an indication of the generalization performance. LOSO is not used for any choice. It only uses the 768 training samples, so the test set stays unseen until the final evaluation.

2 Task 1: Data Pre-processing and Feature Extraction

2.1 Exploratory Data Analysis

The exploratory data analysis has been conducted using only samples from the training set. This prevents leakage from the test set into the processing choices.

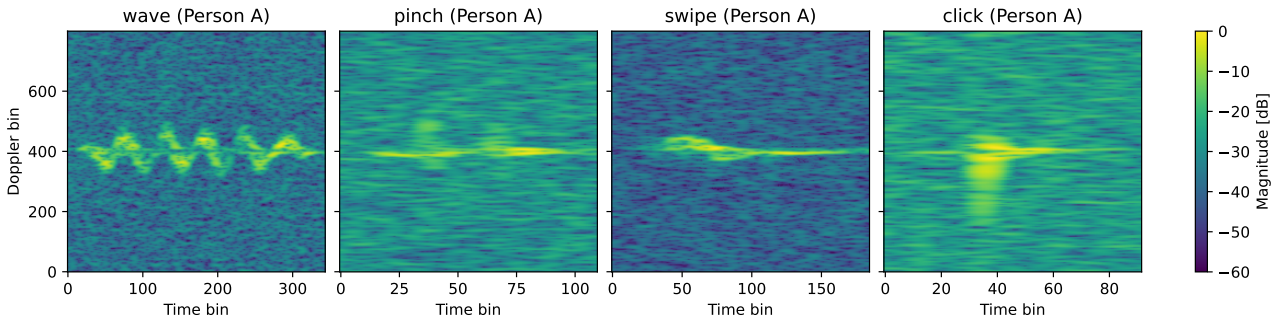


Figure 4: Example micro-Doppler spectrogram for each gesture (magnitude in dB).

The spectrograms all share a similar structure with 800 doppler bins and a varying number of time bins. An example spectrogram of each of the 4 gestures is presented in Figure 4. Zero motion is placed in the centre at doppler bin 400. The spectrograms have already been cropped to the gesture duration, which causes the varying number of time bins between samples. Wave and swipe gestures tend to take longer than click and pinch gestures, as shown in Figure 5. The sample duration therefore holds relevant information that would be removed by resizing all spectrograms to equal length.

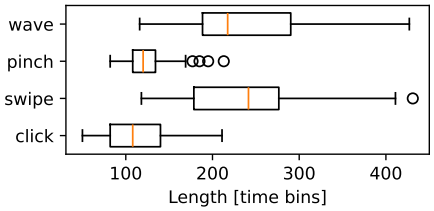


Figure 5: Sample length per gesture.

Yet, the spectrograms have varying magnitude ranges depending on both the gesture and the person. Most of the values however contain background noise instead of gesture energy. The background noise level differs per sample, as seen in Figure 6, making a fixed noise floor for all samples unsuitable.

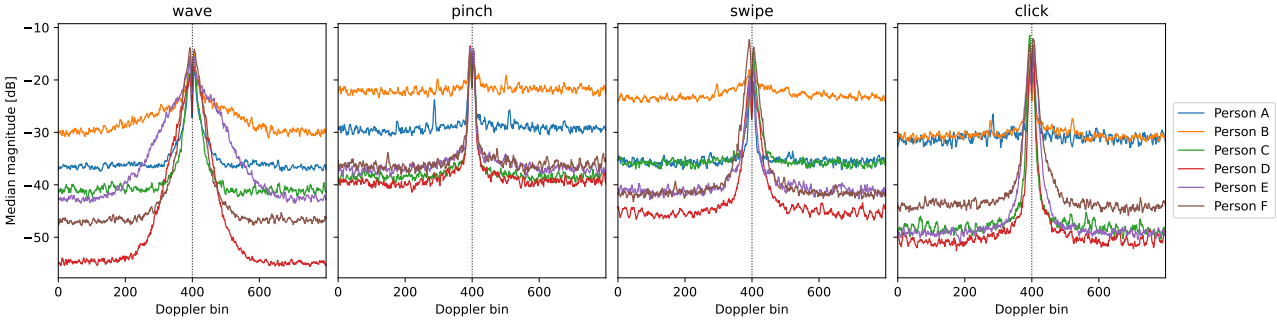


Figure 6: Median magnitude per Doppler bin per gesture and person.

The data needs to be made more uniform by removing noise and rescaling such that the model is prevented from learning a gesture mainly based on the background noise level. This also poses a challenge generalizing the model to more people since the model will learn person specific features with just 6 people in the training data. The gestures should be distinguished on the distribution of gesture energy over time and doppler bins.

2.2 Pre-processing

Since there are no invalid values in the provided dataset, no samples have to be removed. All samples will be processed in four steps: noise removal, cropping, scaling and resizing. The goal of the pre-processing is to turn the raw data into clean spectrograms of uniform dimensions, that can be used for feature extraction and model training.

Noise removal As shown before in Figure 6, the samples all have their own noise floor level. This background noise can be removed since it contains no useful information about the gesture. Because of the sample specific noise floor, the noise removal process also has to be sample specific. Clipping below a fixed threshold for example would cut into the gesture of some samples, while leaving noise in others.

The gesture mainly takes place in the bins close to zero Doppler, while the bins far away from the centre

mainly consist of noise. Figure 6 shows the median levels off to a flat line. By inspection of that same figure, the outer 60 bins on each side are taken as the far bins. The median of the far bin pixels is then defined as the spectrogram's noise level. The median is well suited for this, rather than the mean, since it is less sensitive to a few bins that do contain the gesture (like the broadband stripes of the click gesture). As long as more than half of the far-bin pixels contain noise, the median gives a good estimate of the noise level.

Thresholding at the median would still pass half of the noise samples. Assuming complex Gaussian noise, the power per pixel P is exponentially distributed with mean noise power μ, so the probability that noise exceeds a threshold p is

$$P_{\text{fa}} = \Pr(P > p) = e^{-p/\mu} \;\Rightarrow\; p = \mu \ln(1/P_{\text{fa}}), \tag{1}$$

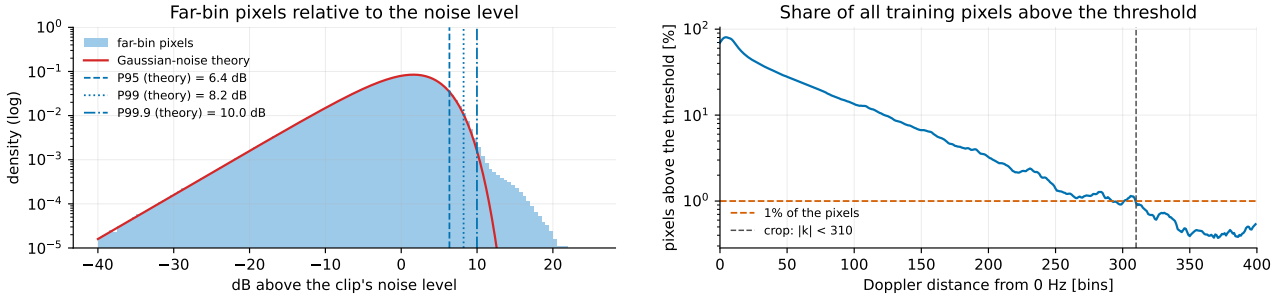
with the median at $p_{50} = \mu \ln 2$. The margin above the median is then independent of the unknown noise power and set only by the chosen false-alarm probability:

$$\Delta_{\text{dB}} = 10 \log_{10}\left(\frac{p}{p_{50}}\right) = 10 \log_{10}\left(\frac{\ln(1/P_{\text{fa}})}{\ln 2}\right). \tag{2}$$

As seen in Figure 7a, the far-bin pixels can indeed be modelled as Gaussian noise. The tail above 10 dB is explained by the broadband stripes in the click gesture. With $P_{\text{fa}} = 10^{-3}$, i.e. the threshold at the 99.9th percentile of the noise, the margin becomes

$$\Delta_{\text{dB}} = 10 \log_{10}\left(\frac{\ln 1000}{\ln 2}\right) \approx 10.0 \text{ dB}. \tag{3}$$

This 99.9% margin is a trade-off between removing more noise and preserving gesture information. For each sample, all values are clipped at the sample specific noise floor plus that fixed 10 dB margin.



(a) Far-bin pixels relative to each sample's noise level, compared with the complex Gaussian noise theory.

(b) Share of all training pixels above the noise threshold per Doppler bin. Bins below 1 % are removed by the crop.

Figure 7: Choice of the noise margin (a) and the Doppler crop (b), derived on the training samples.

Cropping Since the bins far from zero Doppler contain mainly noise and hold little gesture information, they might as well be removed to improve the Doppler resolution when resizing the spectrogram later on. For each Doppler bin, the pixels of all training samples are taken together, and the share that lies above the sample-specific noise threshold is computed (Figure 7b). This share decreases steadily away from zero Doppler. A bin is removed if less than 1% of its pixels exceed the threshold. With a symmetric crop around zero Doppler, this keeps bins 90–709 (620 bins). A narrower crop would give a higher resolution after resizing, at the cost of removing the tails of gestures that do reach into the far bins.

Scaling After the noise removal, each sample still has a different noise floor. The model should not learn the gestures based on this noise floor. The sample dependent noise floor should therefore be converted to a uniform noise floor. This is done using per sample scaling. The noise threshold (median + margin) becomes zero, since it has no information and the maximum (0 dB) becomes 1.

Resizing Finally, the spectrograms have to be made uniform in their dimensions so that they can be used as input for a CNN. A resolution of 128 x 128 is chosen as the output size. The spectrograms are all resized using linear interpolation. This does remove gesture duration, which did hold relevant information so that will be added as a separate feature later on.

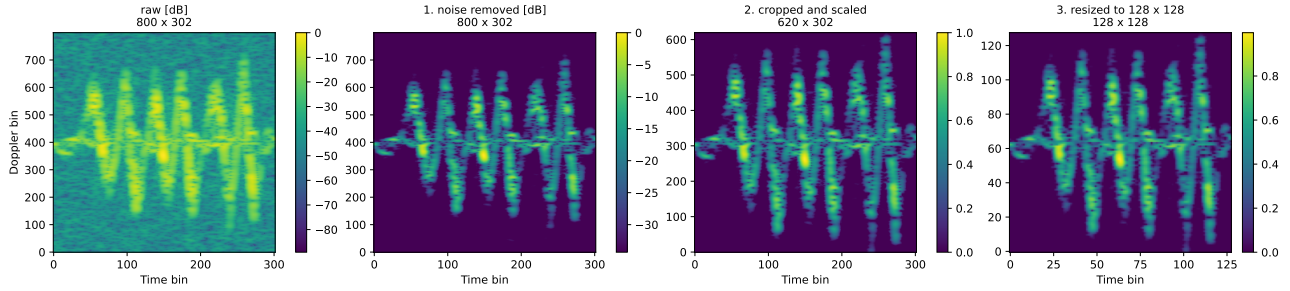


Figure 8: The pre-processing steps applied to a wave sample from the training set.

2.3 Feature Extraction and Selection

The CNN will learn features from the pre-processed and resized spectrograms. The classical models on the other hand would have to process $128^2 = 16384$ pixels, which is a very large number compared to the just 768 available training samples. Resizing also removed the gesture duration. Instead, the feature models will be given a compact representation of the spectrograms based on their micro-Doppler envelopes by Amin et al[2]. All envelope features are computed on the spectrograms after noise has been removed but before and cropping, scaling and resizing so the original duration is preserved.

2.3.1 Envelope extraction

The envelopes are computed on the gesture's linear power above the sample's own noise floor, $P(d, n)$.The noise is removed like in section 2, so that pure noise has power zero. Here d is the Doppler bin relative to zero Doppler (positive is motion towards the radar) and n the time bin. The spectrograms are then cropped in time to just the gesture span: from the first to the last time bin with at least 5% of the power of the strongest time bin.

Effective bandwidth The spectrogram is split into a positive and a negative Doppler half. For each half the gesture power is summed over time and, counting outward from zero Doppler, the effective bandwidth ends where 99.9% of the energy of that half is reached. This Doppler bin is the maximum positive Doppler f_p or maximum negative Doppler f_n of the gesture. A fixed energy share is used as the effective bandwidth. A share of just 99% was found to cut the outer lobes of the gesture, while 99.9% keeps them. Isolated noise cells far from zero Doppler carry almost no energy and stay outside.

Energy and threshold In every time bin the energy of each half is

$$E_U(n) = \sum_{d>0} P(d, n), \qquad E_L(n) = \sum_{d<0} P(d, n), \tag{4}$$

and the threshold is a fixed fraction of it, $T_U(n) = \sigma_U E_U(n)$ and $T_L(n) = \sigma_L E_L(n)$ with $\sigma < 1$. As in the paper, σ is chosen per sample and per half such that the threshold lies exactly at the maximum Doppler in the time bin n_p where that Doppler bin is strongest:

$$\sigma_U = \frac{P(f_p, n_p)}{E_U(n_p)}, \qquad n_p = \arg\max_n P(f_p, n), \tag{5}$$

and likewise σ_L with f_n. In every other time bin the threshold scales with the energy of that time bin.

Envelope The envelope is the outermost Doppler bin that still reaches the threshold:

$$e_U(n) = \max\{d > 0 : P(d, n) \geq T_U(n)\}, \qquad e_L(n) = \min\{d < 0 : P(d, n) \geq T_L(n)\}. \tag{6}$$

Time bins without energy in a half get envelope 0.

Feature vector As in the paper, the envelopes are concatenated into one feature vector, $e = [e_U, e_L]$. Each envelope is placed in a fixed window of 358 time bins, starting at the gesture start and zero-padded after the gesture end. 99% of the training gesture spans fit in this window; the longer samples are cut. Nothing is resampled, so the gesture duration remains visible in the feature vector.

[2]M. G. Amin, Z. Zeng, and T. Shan, "Hand gesture recognition based on radar micro-Doppler signature envelopes," in *Proc. IEEE Radar Conf.*, 2019, pp. 1–6.

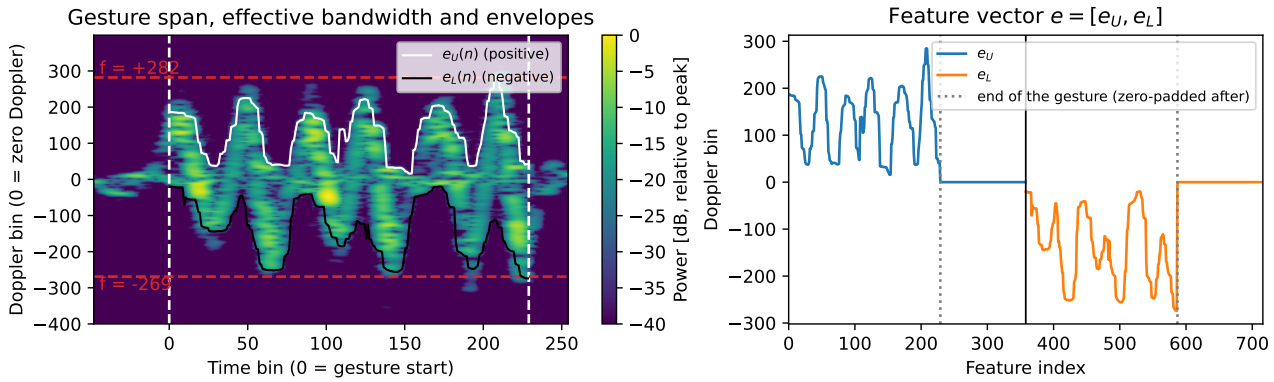


Figure 9: Envelope extraction for one wave sample. Left: gesture power with the gesture span and effective bandwidth (dashed lines) and the envelopes e_U (white) and e_L (black). Right: the resulting zero-padded feature vector $e = [e_U, e_L]$.

Figure 9 shows all steps for a training spectrogram (gesture span, bandwidth, envelopes and resulting feature vector). Figure 10 shows the envelopes for the four different gestures. The envelope length differs between gestures and the extent into Doppler bins also varies. The pinch stays roughly within 60 Doppler bins from the center, while the wave swings further out. The figure also reveals a weak point of the threshold per time bin. For bins that hold little gesture energy, the threshold becomes so low that noise can pass it. As a result, the envelope unintentionally jumps far out.

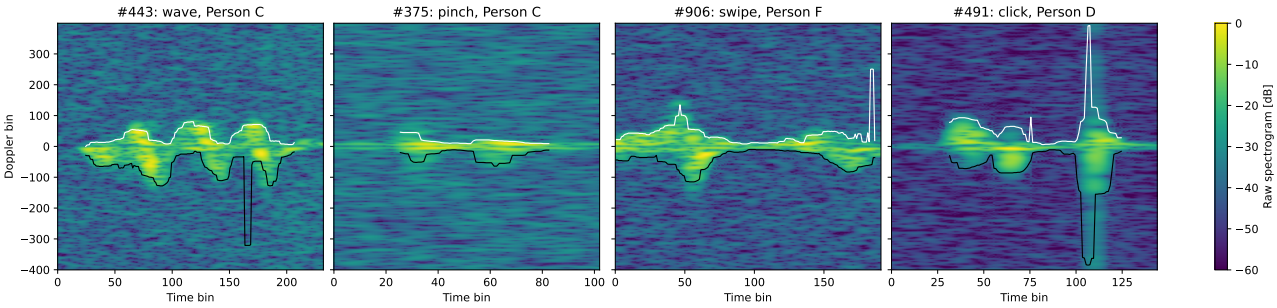


Figure 10: Raw spectrograms with the envelopes e_U (white) and e_L (black) over the gesture span.

2.3.2 Available features

From the envelopes, five feature groups are computed for every sample. Table 1 shows these groups: the first two groups are the features as in the paper by Amin et al., the other three are compact representations of the envelopes for the classical models.

Table 1: Features computed for every sample. e_U and e_L are the positive and negative envelope, f_p and f_n the maximum positive and negative Doppler.

Group	Description	# features
Window envelopes	$e = [e_U, e_L]$, 358 time bins each, zero-padded[2]	716
Empirical features	length T, peak ratio $R = \lvert f_p/f_n \rvert$, bandwidth $B_w = \lvert f_p \rvert + \lvert f_n \rvert^2$	3
Time-normalised envelopes	e_U and e_L, mean over 16 equal segments of the gesture span	32
Envelope statistics	mean, std, median, 90th percentile of $\lvert e_U \rvert$ and $\lvert e_L \rvert$	8
Recording length	length of the whole recording	1
Total		760

The window envelopes keep the full time resolution but with a length of 716 values they are large compared to the number of training samples. Additionally, envelopes of the same gesture performed at different speeds are misaligned. The time-normalized feature set solves this averaging each envelope over 16 equal segments of the gesture span. The gesture duration is kept separately as length T. This also dampens single bin jumps caused by noise. The 90th percentile similarly acts as a robust maximum less sensitive to noise.

3 Task 2: Model Development, Validation and Comparison

3.1 Model Selection and Design

Five models are trained on the pre-processed data: three on the envelope features (Table 1) and two on the 128×128 images.

- **Logistic regression (LR):** multinomial model with the cross-entropy loss and an L2 penalty; the linear, interpretable baseline.
- **SVM:** soft-margin support vector machine with a linear or RBF kernel.
- **Random forest (RF):** decision trees on bootstrap samples with a random subset of features per split, combined by majority vote.
- **SVM + PCA:** PCA reduces the 16 384 pixels, followed by an SVM.
- **CNN:** the deep network, which learns its own features from the images.

The feature models choose between three feature sets: the three empirical features (*empirical*), the empirical features with the time-normalised envelopes, the envelope statistics and the recording length (*compact*, 44 features) and the window envelopes (*window*, 716 features). Scaling and PCA are part of a scikit-learn `Pipeline`, so they are fitted on the training folds only.

3.1.1 CNN layout

The CNN (Table 2) stacks blocks of a 3×3 convolution with ReLU and 2×2 max pooling, followed by dropout and a softmax layer. The convolutions use zero padding, so only the pooling halves the image and the outer pixels, which hold the start and end of a gesture and the highest Doppler velocities, are not lost. ReLU does not saturate for positive inputs, so the gradient does not vanish through the five layers and its sparse output suits the images, in which most pixels are zero after the noise removal. The network is trained with the cross-entropy loss and mini-batch SGD (batch size 32); training stops when the loss on 10% of the training data has not decreased for 10 epochs.

Table 2: Chosen CNN layout (3×3 convolutions, zero padding, ReLU).

Layer	Output shape	Parameters
Input (image)	$128 \times 128 \times 1$	0
Block 1: conv (8) + max pool	$64 \times 64 \times 8$	80
Block 2: conv (16) + max pool	$32 \times 32 \times 16$	1 168
Block 3: conv (32) + max pool	$16 \times 16 \times 32$	4 640
Block 4: conv (32) + max pool	$8 \times 8 \times 32$	9 248
Block 5: conv (32) + max pool	$4 \times 4 \times 32$	9 248
Flatten + dropout (0.25)	512	0
Dense + softmax	4	2 052
Total		26 436

3.2 Validation Procedure and Performance Indicators

The models follow the pipeline of Figure 1 and the split of Figure 3:

1. The model and its hyperparameters are chosen with stratified 5-fold cross-validation on the 768 training samples (validation accuracy); the final model is the model with the highest mean validation accuracy.
2. The chosen setting of every model is evaluated with leave-one-subject-out (LOSO) on the training samples, to see how it generalises to a new person.
3. Every chosen setting is retrained on the entire training set and evaluated once on the 192 test samples, after the final model is chosen.

The classes are balanced, so accuracy is the main metric, together with the macro-averaged precision, recall and F1 score and a confusion matrix. Scores are given as mean ± standard deviation over the folds.

3.3 Hyperparameter Tuning

Every model is tuned with a grid search on 5-fold cross-validation (Table 3); the setting with the highest mean validation accuracy is chosen.

Table 3: Search space and chosen setting (5-fold cross-validation).

Model	Search space	Chosen
LR	feature set; C: 0.01–100	compact, $C = 1$
SVM	feature set; kernel: linear/RBF; C: 0.1–10 000	compact, RBF, $C = 100$
RF	feature set; trees: 100/300/1000; depth: 5/10/20/∞	compact, 300, depth 20
SVM + PCA	components: 5/10/20/50/100; kernel: linear/RBF; C: 0.1–100	20, RBF, $C = 10$
CNN	blocks: 4, 5, 6, …; learning rate: 0.01/0.03/0.1; dropout: 0.25/0.5	5 blocks, 0.03, 0.25

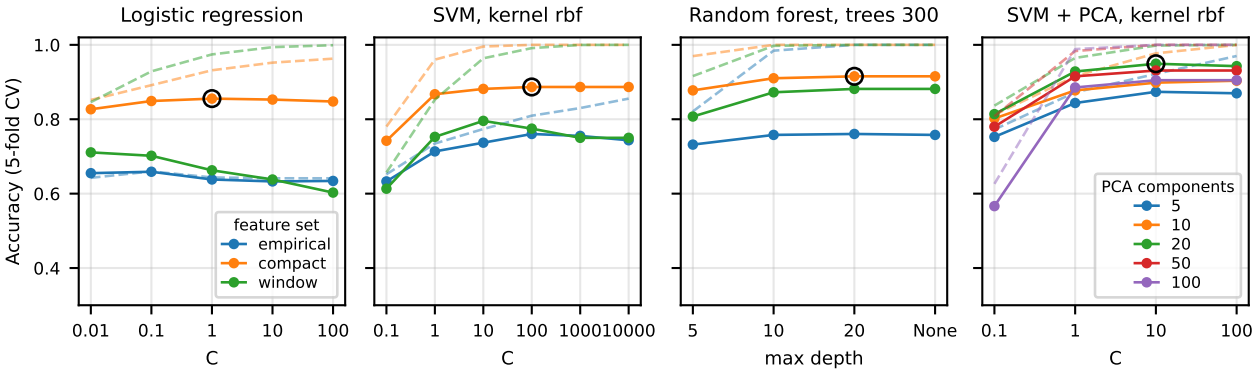


Figure 11: Training (dashed) and validation (solid) accuracy of the classical models; circles: chosen setting.

3.3.1 Classical models

The compact feature set is chosen for all three feature models (Figure 11). The three empirical features alone reach at most 0.76: they describe the size of a gesture, not its course in time. The window envelopes, 716 values for 768 samples, overfit: for LR the training accuracy rises to 1.0 with weaker regularisation while the validation accuracy falls from 0.71 to 0.61. Grids were extended where the chosen value lay at their edge: every SVM $C \geq 100$ gives 0.887 (the training data is fitted perfectly), RF depth 20 equals fully grown trees, 1000 trees are worse than 300 (0.908), and for SVM + PCA 20 components are best (0.87 with 5, 0.90 with 10, 0.93 with 50, 0.91 with 100).

3.3.2 CNN layout search

The number of convolution blocks is chosen first, with the learning rate and dropout fixed at 0.03 and 0.5: starting at four blocks, a block of 32 filters is added until the validation accuracy goes down (Figure 12a). Four blocks reach 0.939, five 0.964 and six 0.961, so five blocks are chosen. The training accuracy is 0.99 for every depth, so the depth changes the generalisation, not the fit. Each block halves the feature map and enlarges the receptive field: with four blocks a unit sees 46 of the 128 pixels, with five blocks 94, which covers most of a gesture, while a sixth block only adds parameters. For five blocks, five of the six learning rate and dropout settings reach 0.96–0.97 (Figure 12b); only learning rate 0.1 with dropout 0.5 is unstable (0.866). Learning rate 0.03 with dropout 0.25 is chosen (0.967).

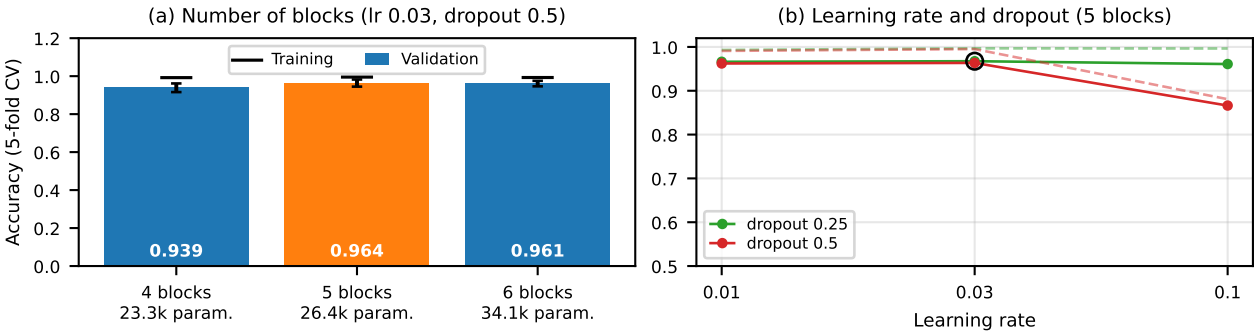


Figure 12: CNN search (5-fold cross-validation): (a) number of blocks, (b) learning rate and dropout.

3.4 Results and Comparison

Table 4: Accuracy of the chosen settings (mean ± std over the folds).

Model	Training	Validation (5-fold)	LOSO	Test	Test F1
LR	0.932	0.855 ± 0.018	0.733 ± 0.055	0.865	0.864
SVM	1.000	0.887 ± 0.022	0.728 ± 0.065	0.875	0.876
RF	1.000	0.915 ± 0.031	$\mathbf{0.747 \pm 0.090}$	0.885	0.886
SVM + PCA	0.998	0.949 ± 0.021	0.612 ± 0.135	**0.953**	**0.954**
CNN	0.997	$\mathbf{0.967 \pm 0.017}$	0.737 ± 0.122	0.948	0.948

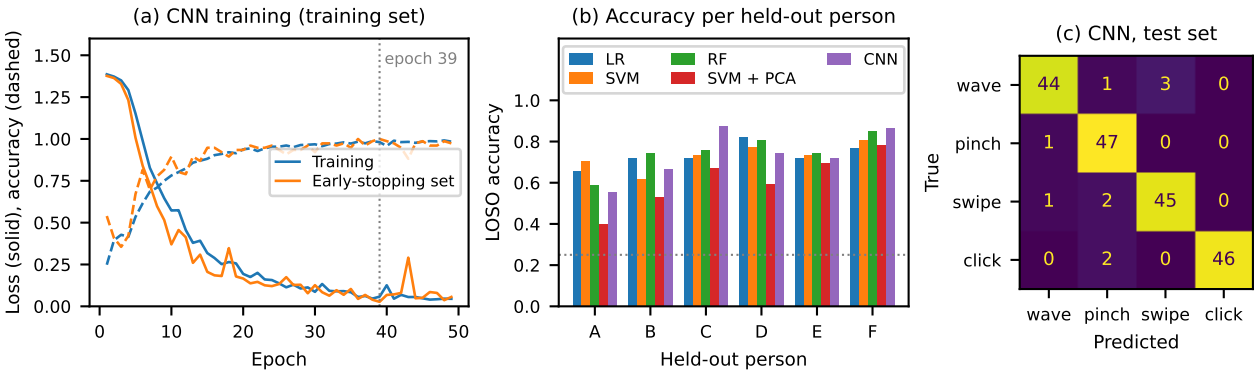


Figure 13: (a) Training of the CNN on the training set, (b) LOSO accuracy per person, (c) confusion matrix of the CNN on the test set.

3.4.1 Model choice

The final model has the highest mean validation accuracy, chosen before the test set is used: the CNN (0.967 ± 0.017, Table 4). Its lead over SVM + PCA (0.949 ± 0.021) is one standard deviation, but on the same folds the CNN is better than SVM + PCA and the RF in four of the five folds (equal to the RF in the fifth) and better than LR and SVM in all five. LOSO supports the choice: for a new person the CNN equals the best feature model and beats SVM + PCA. The RF is the alternative if interpretability or training cost matters (equal under LOSO, 0.05 lower in validation).

3.4.2 Generalisation to a new person

Under LOSO every model loses accuracy (Table 4): the CNN drops from 0.967 to 0.737, SVM + PCA from 0.949 to 0.612 and the feature models to 0.73–0.75. The cross-validation folds contain the same persons as the training folds and therefore reward person-specific details. For a new person the RF, CNN, LR and SVM lie within 0.02, much less than the spread between persons (standard deviation 0.06–0.12); SVM + PCA is clearly the worst, because its components describe where the energy lies in the images of the training persons. Person A is the hardest for most models (Figure 13b).

3.4.3 Test set

The early-stopping loss of the final CNN follows the training loss without rising and is lowest at epoch 39, whose weights are restored (Figure 13a). The test accuracies lie within 0.03 of the validation accuracies: the test set contains the same persons as the training set, so it confirms the cross-validation estimate, not the generalisation to a new person. The CNN reaches 0.948 (macro-F1 0.948). Of its 10 errors, 4 are wave versus swipe and 4 are a swipe or click predicted as pinch (Figure 13c): the long hand movements and the short finger movements. Precision, recall and F1 lie within 0.01 of the accuracy for every model, as expected with balanced classes.

3.4.4 Summary

The CNN is the final model: it has the best validation accuracy and reaches 0.948 on the test set. For a new person all models lose 0.12–0.34 accuracy and the CNN and the feature models end up at the same level (0.73–0.75). In short, more persons are needed to close this gap and to obtain higher accuracies.

4 Conclusions

This work classified four radar hand gestures from micro-Doppler spectrograms. The noise was removed per recording, the Doppler axis cropped and the envelopes of Amin et al. extracted. Five models were chosen with 5-fold cross-validation on the training set, evaluated with leave-one-subject-out (LOSO) and tested once on a held-out test set. A CNN with five convolution blocks on the pre-processed images was the best model: a validation accuracy of 0.967 ± 0.017 and a test accuracy of 0.948 (macro-F1 0.948). Among the feature models the random forest on the compact envelope features was best (0.915). The main limitation is the generalisation to a new person: under LOSO all models lose 0.12–0.34 accuracy, and the CNN (0.74) is not better than the feature models (0.73–0.75). The recordings of one person are highly correlated, and six persons are too few to learn the variation between persons. Future work should add more persons and more recording sessions, vary the gesture speed and timing, and reduce the jumps of the envelopes in weak time bins.

5 AI Acknowledgement

During this assignment, AI has been used. Claude Code was used to write scaffolding code for the plots and refactor prototyping code for better readability. It also helped by adding clear comments to the codebase and was used as a reference to understand python syntax and specific functions. All results were interpreted by the authors, which led to this final report. In the report, AI reviewed drafts versions of sections, which led to suggested improvements that have been evaluated and incorporated where appropriate.